AGT-1: Grav-Residue Calculus, Signed Geometric Channels, and the AGT Program for Signed Gravity-Like Observer Channels

Source-faithful program manuscript

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Abstract

This manuscript is a source-faithful reconstruction of the AGT program line. It separates adopted language, derived structural consequences, imported effective ansatze, target theorems not yet closed, and downstream engineering interpretation. It does not claim closure of the AG-T target chain, experimental confirmation, or theorem evidence from emulator or prototype artifacts.

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1 Protocol and Status Policy

The AGT manuscript is not a detached essay on speculative antigravity. It is the canonical theory-paper translation of the AGT source line, anchored primarily in `docs/Technology/Antigrav_v1.md`, with `Antigrav_Dev_v1.md` and the legacy AGT package tree used only where the source itself crosses into toy-model, prototype, or bridge material. The manuscript therefore follows a literal-transfer discipline first and a stylistic-cleanup discipline second.

1.1 Literal Transfer Rule

The governing rule of the manuscript is source faithfulness. Structural content is translated from the AGT source in source order, with only three allowed normalizations: theorem-style English

prose, consistent notation, and explicit claim-status labeling. The clean manuscript may compress repetitions and remove legacy package chatter, but it may not replace the source theory with a different, tidier one.

1.2 Statement Status Policy

This program adopts a strict separation between adopted language, derived structural consequences, target theorem wrappers, and failure modes.

- Items marked '[A]' are definitions or axioms adopted as part of the AGT program language.
- Items marked '[D]' are derived propositions expected to follow from the imported framework together with AGT-specific adopted language.
- Items marked '[T]' are target theorems. They are theorem-shaped claims that the source program aims to close, but that the present manuscript does not silently upgrade to proved internal results.
- Items marked '[H]' are working hypotheses used conditionally in later sections. They are not silently upgraded to adopted axioms of the AGT core.
- Items marked '[IA]' are imported effective ansatze used explicitly as non-core continuum language.
- Failure modes are mandatory. They are part of the theorem-discipline of the manuscript rather than optional caveats appended after the theory text.

The labels '[H]' and '[IA]' are mandatory whenever later prose relies on imported compatibility or effective-field structure not claimed as internally closed.

In this manuscript, "closed" is reserved only for explicitly qualified formal support claims; it never by itself upgrades an AG-T item to a fully closed physical theorem of the manuscript.

1.3 Imported and Downstream Boundary

The AGT theory line reuses the structural, spectral, observer, and geometric-channel machinery of the main theory without rederiving it. At the same time, it does not treat prototype packages, emulator releases, or hardware-facing notes as theorem evidence. Those downstream lines enter only in the explicit prototype-bridge chapter, where they are named as consequences, dependencies, or acceptance targets rather than as proofs of AGT target statements.

1.4 Source Ledger and Canonical Glossary

The first table records the source-to-manuscript ledger. The second records theorem provenance against source and Lean anchors. The third normalizes the statement statuses and symbols used throughout the text.

AGT Source-to-Manuscript Ledger

This ledger records the source anchors used to assemble the canonical AGT manuscript.

Block	Source lines in Antigrav_v1.md	Canonical manuscript destination	Normalized role
AGT-L1	1–360	Sections 1–2	Project declaration, structural motivation, grav-residue hypothesis, first signed-force formulas, and first program boundary.
AGT-L2	361–789	Sections 1–5	AGT-1 v0.1 protocol, statement statuses, early grav-residue calculus, signed geometric channel, first target theorems, and first toy-model shell.
AGT-L3	790–1303	Sections 3, 4, 6	grav-residue calculus core, neutral composites, dipole refinement, first target theorems AG-T1 and AG-T2, and first chapter-level failure modes.
AGT-L4	1304–1817	Section 5 and downstream microscopic expansion	Microscopic grav-charge, oriented shell defects, composition law, spectral realization of microscopic charge, and target theorems AG-T3–AG-T5.
AGT-L5	1818–2360	Sections 5–6	Field charge densities, mixed-medium closure, screening theory, phase classification, and mesoscopic antigravity target regime AG-T6.
AGT-L6	2361–3406	Sections 6–7	Domain walls, grav-dipoles, antigravity bubbles, neutral shells, and transport sectors for structured backgrounds.
AGT-L7	3407–4027	Section 9	Spectral transport kernel, grav-resolvent, marked moments, branch extraction, and observer-shadow compatibility.
AGT-L8	4028–4630	Section 10	Large-scale mixed-medium consequences, voids, wall networks, and cosmological regime targets.
AGT-L9	4631–14046	Section 11 and downstream artifacts	Observational program, computational pipeline, falsifiability clauses, prototype bridge, and audit boundary.

Convention. "Closed" below never means that the full physical wording of AG-T1–AG-T16 is claimed as a completed theorem of this manuscript. It means only one of the following: (i) a background-conditional Lean closure, or (ii) a source-normalized witness theorem in the synchronized formal layer.

AGT Theorem, Witness, and Formal-Support Provenance Table

Item	Source lines	Lean alignment	Formal support status	Manuscript status
AG5	479–487	`grav_residue_distinct_from_mass_like_spectral_data_of_background`	background-conditional Lean closure	Derived proposition: grav-residue is distinct from mass-like spectral data.
AG11	566–574	`observer_force_link_background` implication chain in `Axioms.lean` and `Theorems.lean`	background-conditional Lean closure	Derived proposition: attraction/repulsion branch law from signed source and probe coupling.
AG17–AG19	1086–1148	`neutral_composite_closure_of_background` and neutral composite witnesses	background-conditional Lean closure	Derived propositions: neutral composite closure and dipole refinement.
AG-T1	586–608 and 1156–1185	target wrapper plus `repulsive_scaling_window_of_background`	background-conditional Lean closure	Target theorem retained in manuscript, though Lean proves the wrapper only under explicit imported background assumptions.
AG-T2	647–659 and 1220–1232	target wrapper plus `spectral_observer_compatibility_of_background`	background-conditional Lean closure	Target theorem retained in manuscript, though Lean proves the wrapper only under explicit imported background assumptions.
AG22–AG43	1357–1674	AGT Lean source block 020–022	partial support	Adopted microscopic language and bridge definitions for integer grav-charge and marked-kernel realization.
AG-T3–AG-T5	1677–1703	`ag_t3_microscopic_charge_witness`, `ag_t4_spectral_sign_matches_microscopic_charge`, `ag_t5_neutral_composite_and_dipole_witness`	source-normalized witness theorem formalized and proved	Manuscript target cluster retained, with formal witness-level support for explicit microscopic sectors, matched spectral sign, and exact neutral-composite plus dipole witnesses.
AG44–AG57 and AG-T6	1842–2146	`ag_t6_mesoscopic_phase_witnesses` in source blocks 030–032	source-normalized witness theorem formalized and proved	Field-regime and screening layer retained as theory target, with formal witness-level support for the four named mesoscopic regimes by explicit phase indicators.

AG67–AG90 and AG-T7–AG-T8	2361–2846	`ag_t7_domain_wall_and_interface_branch_witness`, `ag_t8_interface_sector_witness`	source-normalized witness theorem formalized and proved	Interface-sector language retained as theory target, with formal witness-level support for domain-wall, interface-branch, dipole, neutral-shell, and bubble witnesses.
AG90–AG114 and AG-T9	2847–3406	`ag_t9_transport_class_witnesses`	source-normalized witness theorem formalized and proved	Test-body and background transport layer retained as theory target, with formal witness-level support for bound, repulsive, screened, wall, and bubble transport classes.
AG115–AG142 and AG-T10–AG-T12	3407–4027	AG-T10, AG-T11, and AG-T12 witness theorems in `Theorems.lean`	source-normalized witness theorem formalized and proved	Spectral transport layer retained as theory target, with formal witness-level support for marked-charge matching, wall dictionary data, and a minimal grav-marked kernel plus dominant-branch witness.
AG143–AG169 and AG-T13–AG-T15	4028–4630	AG-T13, AG-T14, and AG-T15 witness theorems in `Theorems.lean` and source block 072	source-normalized witness theorem formalized and proved	Large-scale layer retained as theory target, with formal witness-level support for canonical fluctuation and large-deviation witnesses and for the five named large-scale regimes.
AG170–AG177 and AG-T16	4631–5297	`ag_t16_computational_pipeline_and_audit_witness`	source-normalized witness theorem formalized and proved	Computational schema retained as theory target, with formal witness-level support for explicit pipeline, checkpoint, falsifier, and success-level witnesses.
AGM-1 and later toy line	662–789 and 6227–12236	source blocks 100–117	partial support	Computational and toy-model bridge, not theorem evidence by itself.

AGT Symbol and Status Glossary

This table normalizes the notation and statement-status labels for the AGT program. It is the canonical reference for the clean theory manuscript and for the computational manuscript.

Statement Status Labels

Label	Meaning
[A]	Definition or axiom adopted as part of the AGT program language.
[D]	Derived proposition expected to follow from the imported framework together with AGT-specific axioms.
[T]	Target theorem not yet claimed as proved; used for scaling, compatibility, and existence claims.
[H]	Working hypothesis used conditionally in later sections.
[IA]	Imported effective ansatz used explicitly as non-core continuum language.
F	Failure mode that identifies a concrete collapse mechanism of the program.

Core Symbols

Symbol	Normalized meaning
σ_g	Grav-residue observable with values in $\{-1, 0, +1\}$.
Σ_g	Grav-residue sector attached to the AGT object definition.
ρ_g	Signed observer-level source density induced by grav-residue sectors.
ρ_+, ρ_-, ρ_0	Positive, negative, and neutral coarse sector densities in the mixed-medium regime.
P_g	Grav-polarization order field used for phase, wall, and bubble sectors.
ϕ_g	Signed observer-level potential solving the Green or Poisson channel.
$F_{\mathcal{O}}^{(g)}$	Effective observer-level gravitational force on a probe sector.
$E_{\mathcal{O}}^{(g)}$	Observer-level probe interaction energy in the signed geometric channel.
Γ_g	Positive coupling constant used in the effective probe interaction law.
$K_{\mathcal{O}}^{(g,\mathrm{sgn})}(z, \theta, \eta, s)$	Sign-level grav-marked transport kernel used for the initial spectral formulation of AGT.
$K_{\mathcal{O}}^{(g,\mathrm{int})}(z, \theta, \eta, s)$	Integer-level grav-marked transport kernel used after sharpening to microscopic grav-charge increments.
$Q_{g,\mathrm{sgn}}^{\mathrm{spec}}$	Sign-level spectral grav-charge candidate extracted from the marked dominant branch.
$Q_{g,\mathrm{int}}^{\mathrm{spec}}$	Integer-level spectral grav-charge candidate extracted after sharpening to microscopic grav-charge increments.
τ_g	Wall tension of a grav-domain wall or bubble interface.
ν_0	Neutral fraction of the mixed coarse medium.
Σ_w	Coarse wall-density or wall-measure variable in large-scale regimes.
$n_b(R)$	Bubble-size distribution in the large-scale population description.
μ_{scr}	Screening scale or spectral screening gap in the dressed medium.
$m_\sigma(i)$	Local spectral mass landscape used in the AGT toy model.
$M_\sigma^{\mathrm{in}}(i)$	Local interior spectral quantity reported by the toy-model package.
$Q_{g,\sigma}^{\mathrm{toy}}(i)$	Local spectral grav-charge proxy in the toy-model package.
$U_\sigma(i)$	Effective defect potential used in the 1D scattering realization.
T, R, A	Transmission, reflection, and absorption observables in the scattering experiments.
γ_{int}	Interface-loss parameter controlling the absorption proxy in the resonance experiments.

Canonical AGT Statement Ledger

Item	Status	Canonical interpretation
AG1–AG4	[A]	Grav-residue observable, residue class, rewrite invariance, and transfer stability form the minimal internal AGT language.
AG5, AG11, AG13, AG17	[D]	Grav-residue differs from mass, attraction/repulsion law, and neutral-composite consequences are treated as derived structural consequences.
AG-T1	[T]	Repulsive scaling window theorem: repulsion appears in an observer window with stable signed Green asymptotics.
AG-T2	[T]	Spectral/observer compatibility theorem: spectral grav-charge agrees with observer-level grav-residue sign.
AG123	[H]	Working spectral compatibility hypothesis used conditionally in the later spectral-closure chapter.
AG67–AG86	[IA]	Imported effective phase, interface, and bubble ansatze used for the continuum organization layer.
AG-T6, AG-T8, AG-T9, AG-T12, AG-T15, AG-T16	[T]	Mesoscopic regime, interface-sector, transport, spectral-closure, large-scale regime, and falsifiability targets extend the AGT line beyond the early sign shell.
AG-F1–AG-F5 and AG-F9–AG-F12	F	Failure of residue construction, scaling window, screening closure, coarse mixed-media stability, and interface-sector stability remain mandatory audit items.

The sign-level and integer-level theorem spectral charges $Q^{\mathrm{spec}}_{g,\mathrm{sgn}}$ and $Q^{\mathrm{spec}}_{g,\mathrm{int}}$ must be distinguished from local toy-model proxies such as $Q^{\mathrm{toy}}_{g,\sigma}(i)$; the latter are computational diagnostics and not direct substitutes for theorem-level objects.

Downstream software, emulator, and hardware artifacts are recorded only in the prototype-bridge appendix and are not part of theorem evidence.

2 Imported Foundations and Program Boundary

AGT is a sister microphysical program inside the structural-physical framework, not a replacement for the full book-level theory. The AGT manuscript therefore imports the causal, structural, spectral, and observer layers of the parent framework and then asks a narrower question: can there exist a transfer-stable grav-residue sector whose observer shadow appears as a signed gravity-like channel with repulsive as well as attractive regimes?

2.1 Structural and transfer-compatible observables

The imported structural layer contributes the following fixed discipline.

First, physically meaningful observables are not attached to arbitrary semantic labels, but to stable structural or spectral quantities that survive admissible rewriting and admissible transfer. Second, strict transfer compatibility matters: if an observable is genuinely residue-level or moment-level, it must preserve its numerical shadow under admissible re-encoding rather than change with representation. Third, observer invariance in this framework does not mean independence from every projection choice; it means controlled behavior under the observer and transfer changes that the theory already treats as admissible.

AGT relies on exactly that discipline. Grav-residue is introduced as a new internal observable only because the parent framework already makes room for transfer-stable residue-like observables. Without that imported background, the AGT sign would collapse into a phenomenological label with no structural standing.

2.2 Gravity-like observer channel in the parent theory

The imported geometric layer is equally important. In the parent theory, gravity-like behavior appears only after projection to an observer-accessible geometric regime. A state is first coarse-grained or projected to a graph or continuum surrogate, then a Green- or Poisson-type channel is defined on that surrogate, and only after that does one read observer-level potential and effective force. The force is therefore not a primitive of the underlying theory; it is an observer-level shadow of a deeper structural channel.

AGT does not alter that ordering. It keeps the same projected geometric channel but replaces the unsigned source by a signed source built from grav-residue. The resulting AGT channel is still observer-level, still projection-dependent, and still meaningful only inside a stable scaling window. What changes is the sign structure of the source, not the methodological status of the channel.

2.3 Program boundary

The AGT program keeps an explicit claims boundary.

It does not assume negative inertial mass. In the parent framework, inertial mass is read spectrally from transport branches, so AGT does not replace that machinery with a sign flip in Newtonian language. It does not claim exact Newtonian closure a priori; any $1/r^2$ regime must arise only inside a stable scaling window of the projected geometric channel. It does not treat toy-model output, emulator output, or prototype packages as proof of the AGT target theorem chain. And it does not claim that every observer window must see the same antigravity regime; as elsewhere in the framework, the observer shadow is a derived regime, not a metaphysical primitive.

2.4 Imported objects and standing hypotheses

Later sections use imported objects and standing hypotheses from the parent framework and from explicitly marked effective layers. To keep those imports visible, we fix the following standing list.

IH1. Structural state space W and the notion of stable sectors or stable structural classes are available.

IH2. An admissible observer tuple $\mathcal{O} = (\Pi_{\mathcal{O}}, \Theta_{\mathcal{O}}, \chi_{\mathcal{O}})$ exists on the working window.

IH3. The observer projection induces an admissible graph or continuum geometry on the working window.

IH4. A Green- or Poisson-type operator exists on that projected geometry, together with the corresponding potential solver.

IH5. An admissible observer gradient or equivalent force-reading operation exists whenever an effective force law is written.

IH6. The marked-kernel construction admits an admissible dominant branch or equivalent on-shell singular structure on the working spectral window.

IH7. A low-energy transport expansion exists whenever rest energy or inertial amplitude is extracted from a branch.

IH8. Whenever continuum order fields, Yukawa screening, kink profiles, or bubble energies are used, they enter only as imported effective ansatze rather than as closed derivations from the AGT core.

IH9. A coarse-graining window exists whenever field densities, hydrodynamic closures, or large-scale mixed-media variables are introduced.

IH10. Replay, rewrite, and strict-transfer stability remain the relevant notions of representational admissibility throughout the AGT program.

Unless stated otherwise, later sections cite only the subset of IH1–IH10 actually used in the displayed constructions.

3 Grav-Residue Calculus

This chapter introduces the first internal AGT language: grav-residue values, grav-residue classes, rewrite and transfer stability, and the explicit separation between grav-residue and mass-like spectral data. These statements are source-faithful translations of the early AGT project core and are synchronized with the first Lean AGT language objects in `GravResidue.lean`.

3.1 Definitions and axioms

Definition 3.1 [A]. AG1. Grav-residue values and observable. Define the grav-residue value set

$$\Sigma_g := \{-1, 0, +1\}.$$

An AGT grav-residue observable is a map

$$\sigma_g : W \to \Sigma_g$$

on structural states, interpreted as ordinary gravitating, antigravitating, and grav-neutral values respectively.

Definition 3.2 [A]. AG2. Grav-residue class. If $\mathfrak{p}$ is a stable EFU- or assembly-class, its grav-residue class is the value

$$\sigma_g(\mathfrak{p}).$$

The role of the class is exactly the same as for other residue observables in the parent framework: it is meaningful only once the state has been promoted from raw representation to a stable physical sector.

Definition 3.3 [A]. AG3–AG4. Rewrite and transfer stability. The AGT source fixes two adopted invariance clauses.

First, if

$$x \to_R y$$

inside one admissible local rewriting class, then

$$\sigma_g(x) = \sigma_g(y).$$

Second, if two representations are related by admissible strict transfer or admissible simulation-preserving re-encoding, then the numerical grav-residue shadow is preserved. Grav-residue is therefore introduced as a transfer-stable internal observable rather than as an observer-facing label.

3.2 Derived propositions

Proposition 3.4 [D]. AG5. Grav-residue is distinct from mass-like spectral data. *The grav-residue sign is not identified with observer-level mass, total energy, or inertial mass amplitude. In AGT, grav-residue is an internal residue observable, whereas mass-like quantities remain spectral transport quantities extracted from dominant branches, poles, or low-energy transport expansions.*

Lean-backed proof sketch under the imported background. This separation is exactly the one formalized by the Lean theorem `grav_residue_distinct_from_mass_like_spectral_data_of_background`. Under the imported spectral-mass-separation background, grav-residue enters as an internal residue observable while mass-like quantities are read from the spectral transport layer. A human-readable translation of that closure is the following: the sign that controls entry into the signed observer channel cannot be the same object as the mass-like data extracted from low-energy spectral branches, because the AGT source explicitly rejects the negative-mass collapse.

Dependency status. This item is Lean-closed relative to the imported background theorem `spectral_mass_separation_background`; it is not claimed as a source-internal proof independent of those imported hypotheses.

Proposition 3.5 [D]. AG6. Grav-residue is a candidate observer-invariant internal observable. *If grav-residue is realized by a transfer-compatible residue or marked-moment construction, then it is a natural candidate for observer-invariant internal data in the sense relevant to the structural-physical framework.*

Conditional proof sketch under the imported background and standing hypotheses. By IH1, IH6, and IH10, the parent framework already treats transfer-compatible residue observables and marked-kernel observables as the right layer from which observer-invariant internal data may be read. AGT does not add a new invariance doctrine; it reuses that one. Hence once σ_g is realized at the residue or marked-moment level, it inherits the same observer-compatibility expectation.

Dependency status. This item is not claimed as a closed source-internal proof unless the imported observer- and transfer-compatibility machinery is made explicit.

4 Signed Geometric Channel and Effective Force

This chapter lifts grav-residue into the observer-level geometric channel. It gives the signed source law, the signed Green or Poisson channel, the probe interaction energy, and the first attraction/repulsion law. The content is source-faithful to the early AGT shell and remains deliberately observer-level rather than fundamental.

4.1 Observer-level source construction

Definition 4.1 [A]. AG7. Signed source density. For a coarse-grained family of stable objects, define the signed source density

$$\rho_g(v) := \sum_a \sigma_g(a)\, \mu_a(v), \qquad \mu_a(v) \geq 0.$$

The source is therefore not inserted by hand as an external minus sign in a field equation. It is built from the distribution of grav-residue sectors themselves.

Definition 4.2 [A]. AG8. Signed Green channel. The AGT potential is defined by the same observer-level geometric operator as in the parent gravity-like channel,

$$(L + \varepsilon I)\phi_g = \rho_g.$$

AGT changes the sign structure of the source, not the methodological status of the channel.

Definition 4.3 [A]. AG9–AG10. Observer potential, probe interaction energy, and effective force. For a fixed observer $\mathcal{O} = (\Pi_{\mathcal{O}}, \Theta_{\mathcal{O}}, \chi_{\mathcal{O}})$, define the observer-level potential by

$$V_{\mathcal{O}}^{(g)} = \chi_{\mathcal{O}}(\phi_g).$$

For a probe sector p, define the interaction energy

$$E_{\mathcal{O}}^{(g)}(p \mid x) = -\Gamma_g \, \sigma_g(p) \, V_{\mathcal{O}}^{(g)}(x), \qquad \Gamma_g > 0,$$

and, whenever the observer-space gradient is available, the effective force

$$F_{\mathcal{O}}^{(g)}(p \mid x) = -\nabla_{\mathcal{O}} E_{\mathcal{O}}^{(g)}(p \mid x).$$

This matches the parent theory discipline exactly: the force is the observer-level response of a distinguished probe sector to the projected geometric landscape.

4.2 Two-sector interaction law

Assume under IH2–IH5 that a localized source sector A with grav-residue σ_A produces, in some observer scaling window, a leading potential of the form

$$\phi_A(r) \approx \kappa \sigma_A G_*(r), \qquad \kappa > 0, \qquad G_*'(r) < 0 \text{ on the working window.}$$

Then a probe sector B with grav-residue σ_B has interaction energy

$$U_{AB}(r) = -\Gamma_g \sigma_B \phi_A(r) \approx -\Gamma_g \kappa \sigma_A \sigma_B G_*(r).$$

Differentiating radially gives the effective force law

$$F_{AB}(r) \approx \Gamma_g \kappa \sigma_A \sigma_B \partial_r G_*(r) \, \hat{r}.$$

Proposition 4.4 [D]. AG11. Attraction/repulsion law. *If $\sigma_A \sigma_B > 0$, the force is attractive. If $\sigma_A \sigma_B < 0$, the force is repulsive. Antigravity therefore appears as the repulsive branch of one and the same signed observer-level geometric channel.*

Lean-backed proof sketch under the imported background. The displayed formula is the human-readable observer-channel shadow of the Lean background implication `observer_force_link_background_implies_repulsive_scaling_window`. Once IH2–IH5 hold and the working kernel admits the relevant observer-force background, the sign of the force is controlled entirely by the product $\sigma_A \sigma_B$: positive product yields the attractive branch and negative product yields the repulsive branch. No second force law is introduced; the sign flip is already encoded in the signed source and probe coupling.

Dependency status. This item is Lean-closed relative to the imported background `observer_force_link_background`; it is not claimed as a source-internal proof with no imported observer-channel assumptions.

Corollary 4.5 [D]. Special three-dimensional Coulomb/Newton window. *If the working observer window specializes to the three-dimensional Coulomb or Newton regime with $G_*(r) = 1/r$, then the standard $1/r^2$ sign law is recovered.*

5 Spectral Formulation and Compatibility Targets

This chapter records the spectral side of the AGT program after the early sign-only shell. It now includes the source transition from sign-valued grav-residue to integer-valued microscopic grav-charge, the oriented-shell defect model, and the grav-marked transport kernel that is meant to carry the microscopic charge into spectral language. At manuscript level AG-T2 remains a target theorem, while the synchronized Lean layer now proves source-normalized witness versions of the later microscopic cluster AG-T3 through AG-T5.

5.1 Microscopic grav-charge and oriented shell defects

The AGT source strengthens the early sign-valued shell by introducing an integer microscopic grav-charge

$$q_g : W \to \mathbb{Z}, \qquad \sigma_g(x) = \operatorname{sgn}(q_g(x)).$$

This strengthening matters for two reasons. First, it distinguishes exact neutrality from partial cancellation. Second, it creates room for dipole and higher multipole moments without collapsing everything to a ternary sign label.

The preferred first microscopic model in the source is an oriented-shell defect model. A normalized structural representative is decomposed into local blocks u, each carrying an orientation sign $\varepsilon(u) \in \{-1, +1\}$ and a nonnegative shell load $\nu(u)$. The microscopic grav-charge is then read as an oriented shell sum,

$$q_g(x) = \sum_{u \in H(x)} \varepsilon(u)\, \nu(u),$$

or equivalently through the corresponding canonical representative $c(x)$ when that normalization is fixed. The key point is that grav-charge is now structural and integer-valued before any observer projection is taken.

The source also fixes the microscopic invariance conditions that must hold if this definition is to count as physical. The grav-charge must be stable under the relevant h-equivalence, preserved under admissible local rewriting, stable under admissible strict transfer, and nontrivial on at least one positive, one negative, and one neutral class. Those conditions are exactly what later target theorems AG-T3 through AG-T5 are meant to close.

5.2 Marked-kernel construction

Definition 5.1 [A]. AG12. Grav-marked transport kernel. For a fixed observer $\mathcal{O}$, define the grav-marked transport kernel

$$K_{\mathcal{O}}^{(g,\mathrm{sgn})}(z, \theta, \eta, s) = \sum_e w(e) z^{\gamma(e)} e^{i\theta \cdot \Delta x_{\mathcal{O}}(e)} e^{-\eta a(e)} s^{\delta \sigma_g(e)} [[e]].$$

The new marker s tracks grav-residue flow in exactly the same way that the parent framework uses marked kernels to extract other transport and moment observables.

Definition 5.2 [A]. AG13. Spectral grav-charge. If $\lambda_*(z,\theta,\eta,s)$ is the dominant branch of the marked kernel, define the sign-level spectral grav-charge candidate by

$$Q_{g,\mathrm{sgn}}^{\mathrm{spec}} \sim \partial_{\log s} \log \lambda_* \big|_{s=1}.$$

This is the spectral candidate for the observer shadow of grav-residue.

The microscopic source then sharpens this construction by attaching increments $\Delta q_g(e) \in \mathbb{Z}$ to elementary events. In that sharpened form the grav-marked kernel reads

$$K_{\mathcal{O}}^{(g,\mathrm{int})}(z,\theta,\eta,s) = \sum_e w(e) z^{\gamma(e)} e^{i\theta\cdot\Delta x_{\mathcal{O}}(e)} e^{-\eta a(e)} s^{\Delta q_g(e)} [[e]],$$

and the resulting marked moment

$$Q_{g,\mathrm{int}}^{\mathrm{spec}} \sim \partial_{\log s} \log \lambda_* \big|_{s=1}$$

becomes the spectral image of the integer grav-charge rather than merely the sign-valued shell.

5.3 Compatibility target theorem

Theorem 5.3 [T]. AG-T2. Spectral/observer compatibility. *The AGT source targets a compatibility theorem of the following form: for stable sectors, the sign of the spectral grav-charge agrees with the observer-level grav-residue sign,*

$$\operatorname{sgn} Q_{g,\mathrm{int}}^{\mathrm{spec}} = \sigma_g.$$

In a stronger form, one seeks a positive scale factor C_g such that $Q_{g,\mathrm{int}}^{\mathrm{spec}} = C_g q_g$ at least on low-energy stable sectors.

This theorem is recorded as a target wrapper rather than as a proved internal result. That status remains consistent both with the AGT source and with the Lean package: the full observer/spectral compatibility theorem still lives in the target suite, even though Lean now also records a source-normalized sign-matching witness once the dominant-branch grav slope is identified with microscopic grav-charge.

The same source block also isolates three further targets. AG-T3 asks for a microscopic sign theorem: the oriented-shell construction should produce a transfer-stable internal grav-charge. AG-T4 is the sharpened spectral compatibility theorem identifying the sign of the marked spectral quantity with the sign of the microscopic charge. AG-T5 is the neutral-composite target theorem: there should exist a nonempty class of exact grav-neutral composites whose leading monopole grav-charge vanishes while higher multipole response remains available. The synchronized Lean layer now proves source-normalized witness versions of all three targets: explicit positive, negative, and neutral microscopic sectors; sign agreement between microscopic charge and a matched spectral branch; and an exact neutral composite together with a nontrivial dipole witness.

6 Toy Model, Field Regimes, and Scaling Targets

This chapter performs three source-faithful jobs at once. First, it records the early repulsive-scaling shell and the first toy-model bridge AGM-1. Second, it carries in the later source field-regime material in which integer grav-charge is coarse-grained into charge densities, neutral fractions, screening media, and segregated phases. Third, it makes explicit that AG-T1 and AG-T6 are not competing theorems, but the microscopic and mesoscopic ends of the same program.

6.1 Field charge density and signed medium

Definition 6.1 [A]. AG44–AG48. Coarse signed density fields. On a coarse observer graph or coarse continuum window one introduces charged densities

$$n_k^+(x),\ n_k^-(x),\ n_k^0(x).$$

Here $n_k^{\pm}(x)$ are number densities of sectors carrying microscopic grav-charge $\pm k$, while $n_k^0(x)$ denotes neutral-sector number density. The signed coarse grav-charge density is then defined by

$$\rho_g(x) = \sum_{k\geq 1} k\, n_k^+(x) - \sum_{k\geq 1} k\, n_k^-(x).$$

One simultaneously retains a separate positive mass-like density field. This separation is essential: a grav-neutral medium may still be massive, and a massive medium may still be screened in the grav-channel.

The full signed channel is then read in the same projected form as before,

$$(L_{\text{eff}} + \varepsilon)\phi_g = \rho_g,$$

or, in isotropic continuum notation,

$$(\kappa\Delta + \varepsilon)\phi_g = \rho_g.$$

This is the point where AGT becomes a theory of mixed media rather than only a two-body sign rule.

Definition 6.2 [A]. AG49–AG53. Polarization and neutral fractions. The later field-regime source also introduces the coarse positive, negative, and neutral densities

$$\rho_+(x), \qquad \rho_-(x), \qquad \rho_0(x),$$

the grav-polarization field

$$P_g(x) = \rho_+(x) - \rho_-(x),$$

and the neutral fraction

$$\nu_0(x) = \frac{\rho_0(x)}{\rho_+(x) + \rho_-(x) + \rho_0(x) + \delta_0}.$$

These are the minimal mesoscopic variables needed to distinguish pure charged media, screened mixtures, and neutral-composite-rich regimes.

6.2 Schematic mixed-medium closure ansatz

The source does not yet claim a closed continuum derivation, but it does write the first hydrodynamic shell for mixed grav-media. In normalized notation the three-sector continuity block takes the form

$$\partial_t\rho_+ + \nabla\cdot(\rho_+ u_+) = -\Gamma_b + \Gamma_u,$$
$$\partial_t\rho_- + \nabla\cdot(\rho_- u_-) = -\Gamma_b + \Gamma_u,$$
$$\partial_t\rho_0 + \nabla\cdot(\rho_0 u_0) = 2\Gamma_b - 2\Gamma_u,$$

where Γ_b is the binding rate into neutral composites and Γ_u is the unbinding rate back into free charged sectors.

The conversion terms are chosen so that the signed coarse charge is conserved on the working window:

$$\partial_t \rho_g + \nabla \cdot J_g = 0, \qquad J_g := \sum_{k\geq 1} k\, n_k^+ u_k^+ - \sum_{k\geq 1} k\, n_k^- u_k^-.$$

The matching momentum-type closure reads schematically as

$$\rho_\pm(\partial_t u_\pm + u_\pm \cdot \nabla u_\pm) = -\nabla p_\pm \pm \Gamma_g \rho_\pm \nabla \phi_g + \mathcal{D}_\pm + \mathcal{I}_\pm,$$

while the neutral block carries no leading signed-force term and couples only through conversion, diffusion, and induced higher responses. This is the first point at which screening, phase separation, and repulsive drift enter AGT as competing mesoscopic processes rather than only as microscopic slogans.

Proposition 6.3 [D]. AG54–AG57. Mesoscopic regime separation. *Assume IH4, IH8, and IH9, and assume that the binding-unbinding competition does not collapse trivially on the working coarse window. Then the AGT medium admits at least four physically distinct observer regimes: an unscreened attractive regime, a mixed repulsive regime, a screening regime rich in neutral composites, and a segregated regime with large positive and negative grav-domains.*

Lean-backed witness summary. The synchronized Lean layer now records the witness theorem `ag\_t6\_mesoscopic\_phase\_witnesses`. It exhibits explicit phase-indicator data realizing the four named regimes: unscreened attraction, mixed repulsion, screening, and segregated antigravity. This does not replace the imported continuum closure used in the prose statement, but it does provide source-normalized witness-level support for the regime-separation claim at the level of concrete witnesses rather than leaving it as a purely verbal target.

Dependency status. Lean proves a source-normalized witness version of the four regimes. The full observer-continuum reading in the manuscript still depends on the imported coarse-graining and closure hypotheses.

6.3 Open-chain strip and bubble realizations

The AGT source introduces a first toy-model shell in which grav-residue sectors are represented on a finite, computable reduction before being transferred to the later v1.0 and v1.1x numerical packages. In the current recorded workspace, that line survives as a downstream numerical package together with associated strip and bubble resonance reports.

At theory level the toy-model bridge serves three purposes. It provides a finite realization in which opposite-sign sectors can be distinguished. It supplies explicit observables for transport through inhomogeneous backgrounds. And it gives the first numerical semantics for transmission, reflection, and absorption-type observables,

$$T, \qquad R, \qquad A,$$

which later become the basic outcome coordinates for wall, bubble, and resonance scans.

The source is careful here: the toy line is not theorem evidence by itself. It is an admissible finite reduction on which one can check whether the signed source law, the mixed-medium closure, and the interface observables survive a computational projection.

6.4 Repulsive scaling window theorem

Theorem 6.4 [T]. AG-T1. Repulsive scaling window. *Suppose that the observer projection produces a stable power-law or Coulomb-like Green regime, that grav-residue is transfer-stable, and that the signed source law remains well defined on the working observer window. Then there exists a scaling window in which opposite-sign grav-residue sectors exhibit repulsive force behavior, while same-sign sectors exhibit attractive behavior.*

The theorem is a target theorem, not a claimed theorem under explicit imported background closure. Its role is to state the first macroscopic physical goal of the AGT line in exact form.

The source also records a first phase-regime decomposition that remains useful in the clean manuscript. The four earliest regimes are pure attractive sectors, mixed repulsive sectors, screening regimes, and runaway failure regimes. The later toy-model line refines these into explicit strip, wall, bubble, quasi-capture, and resonance windows, but the early phase picture is already present in the source shell and is the correct theory-level precursor to the numerical reports.

Theorem 6.5 [T]. AG-T6. Mesoscopic mixed-medium regime theorem. *Suppose that microscopic grav-charge is transfer-stable, that free and bound opposite-sign sectors coexist, and that IH4, IH8, and IH9 hold on the working coarse window. Then the AGT medium admits at least four mesoscopic regimes: unscreened attraction, mixed repulsion, screening by neutral composites, and segregated antigravity domains.*

AG-T6 belongs here because it is the large-scale completion of the repulsive-scaling program started by AG-T1. AG-T1 says that repulsion can exist in a stable observer window. AG-T6 says that, once one allows coarse densities, conversion rates, and mixed-media closures, repulsion becomes only one phase among several sector-level macroscopic regimes.

7 Composite Sectors, Interfaces, Screening, and Failure Modes

This chapter collects the composite closure and interface logic of the AGT line once both the early sign shell and the later field-regime shell are taken seriously. It records the minimal neutral-composite ansatz, the dipole refinement, the screening logic of mixed media, the first order-parameter language for grav-phases, and the later source block on domain walls, shells, bubbles, and interface-localized sectors.

7.1 Composite grav-residue closure

Definition 7.1 [A]. AG14–AG16. Composite grav-residue and minimal closure ansatz. If $\mathfrak{a}$ and $\mathfrak{b}$ are stable sectors, their composite grav-sector is written

$$\mathfrak{c} = \mathfrak{a} \otimes \mathfrak{b}.$$

At the sign-only level, the source introduces a first effective closure law by a composition map

$$\sigma_g^{\mathrm{eff}}(\mathfrak{a} \otimes \mathfrak{b}) = \mathcal{C}_g\big(\sigma_g(\mathfrak{a}), \sigma_g(\mathfrak{b}), \Delta_g(\mathfrak{a}, \mathfrak{b})\big),$$

and then specializes to a minimal closure ansatz in which opposite-sign equal-strength sectors neutralize to zero at leading monopole order.

Proposition 7.2 [D]. AG17. Grav-neutral composites. *Under the minimal closure ansatz, an opposite-sign pair with equal and opposite grav-residue produces a grav-neutral composite. The resulting composite can remain massive or structurally nontrivial while contributing zero leading monopole grav-charge.*

Lean-backed proof sketch under the imported background. This is the human-readable form of the Lean closure packaged by `neutral_composite_closure_of_background`. Once the imported neutral-composite background is assumed, equal and opposite constituent residues cancel at leading monopole order, while nothing in that closure forces the composite itself to disappear or lose inertial content.

Dependency status. This item is Lean-closed relative to the imported background theorem `neutral_composite_closure_background`; it is not claimed as a source-internal proof independent of that background.

Proposition 7.3 [D]. AG18–AG19. Dipole refinement of grav-neutral composites. *Even when the leading composite grav-charge vanishes, a nonzero grav-dipole moment can remain. In that case the monopole contribution disappears but a subleading dipolar field remains available as the first nonzero far-field term.*

Conditional proof sketch under the imported background and standing hypotheses. Assume IH3, IH4, and IH9. The source introduces this refinement precisely to avoid confusing neutrality with complete grav-invisibility. Once positive and negative constituents occupy distinct positions, the signed first spatial moment need not vanish even if the total charge does. The first surviving far-field term is then dipolar rather than monopolar.

Dependency status. This item is not claimed as a closed source-internal proof unless the multipole expansion and coarse field hypotheses are made explicit.

Once the later field shell is included, this same logic becomes the basis of screening. Opposite-sign constituents can either remain free, in which case the medium continues to carry large-scale signed grav-charge, or bind into neutral composites, in which case leading monopole charge is removed from the long-range channel. The AGT source therefore treats neutral composites not as an optional side remark, but as the bridge between microscopic opposite-sign content and macroscopic suppression of the grav-channel.

7.2 Grav-phase order parameter and wall background

This subsection uses a standard imported effective-field ansatz for phase and interface organization. It is not presented as an internally closed derivation from the AGT core.

Imported effective ansatz 7.4 [IA]. AG67–AG69. Grav-phase order field and minimal potential. The coarse order field of the mixed medium is the grav-polarization

$$P_g(x) = \rho_+(x) - \rho_-(x).$$

Its phase structure is modeled by an effective potential $V_{\text{eff}}(P_g)$ with stable minima corresponding to macroscopically favored positive or negative grav-sectors. The minimal explicit realization is the double-well ansatz

$$V_{\text{eff}}(P_g) = \frac{a}{4}(P_g^2 - P_0^2)^2, \qquad a > 0, \quad P_0 > 0,$$

with degenerate minima at $P_g = \pm P_0$.

This is the source-faithful point at which AGT becomes a genuine phase theory. Positive and negative sectors are no longer only local sign labels. They are now candidate macrophases of the mixed medium.

Imported interface-energy ansatz 7.5 [IA]. AG70–AG75. Domain walls and wall tension. A grav-domain wall is a configuration in which $P_g(x)$ interpolates between opposite stable minima, for example

$$P_g(x) \to +P_0 \quad \text{as } x \to -\infty, \qquad P_g(x) \to -P_0 \quad \text{as } x \to +\infty.$$

The corresponding interface functional is

$$\mathcal{E}[P_g] = \int dx \left[\frac{\kappa_g}{2} (\partial_x P_g)^2 + V_{\text{eff}}(P_g) \right],$$

and the wall tension is

$$\tau_g = \int_{-\infty}^{\infty} dx \left[\frac{\kappa_g}{2} (P_g')^2 + V_{\text{eff}}(P_g) - V_{\min} \right].$$

Proposition 7.6 [D]. AG72–AG75. Kink profile and interface cost. *Under the ansatz of Imported effective ansatz 7.4 and Imported interface-energy ansatz 7.5, the minimizing wall profile is of kink type,*

$$P_g(x) = P_0 \tanh\left(\frac{x - x_0}{\ell_g} \right), \qquad \ell_g \sim \sqrt{\frac{\kappa_g}{a P_0^2}},$$

and every stable wall has positive tension $\tau_g > 0$.

Conditional proof sketch under the imported background and standing hypotheses. Assume IH8 and IH9. The stationary Euler-Lagrange equation of the imported interface functional is

$$\kappa_g P_g'' = a P_g (P_g^2 - P_0^2),$$

whose finite-energy heteroclinic solution is the standard kink profile. Positivity of the wall tension follows because the integrand vanishes only in the homogeneous minima and is nonnegative for the interpolating solution after subtraction of the bulk vacuum value.

Dependency status. This item is not claimed as a closed source-internal proof unless the imported effective-field ansatz is accepted on the stated working window.

7.3 Grav-dipoles, neutral shells, and bubbles

Definition 7.7 [A]. AG76–AG80. Dipoles and neutral shells. A grav-dipole is a localized composite sector with vanishing total grav-charge but nonzero signed first spatial moment,

$$Q_g^{\text{tot}} = \int \rho_g(x)\, dx = 0, \qquad \mathbf{d}_g = \int x\, \rho_g(x)\, dx \neq 0.$$

A neutral shell is a composite configuration in which an inner region and an outer shell carry opposite net grav-charge so that the total monopole charge is small or vanishes at leading order.

Proposition 7.8 [D]. AG78–AG80. Subleading far-field structure. *The far field of a grav-dipole is dipolar rather than monopolar,*

$$\phi_g(r) \sim \frac{\mathbf{d}_g \cdot \hat{r}}{r^2}, \qquad F_g(r) \sim \frac{1}{r^3},$$

and a sufficiently symmetric neutral shell suppresses the long-range field more strongly than the corresponding bare charged core.

Conditional proof sketch under the imported background and standing hypotheses. Assume IH3, IH4, IH8, and IH9. These are the standard first nonvanishing multipole consequences once the monopole term cancels. The source uses them to separate neutrality from triviality: long-range response is suppressed, but higher grav-response remains available.

Dependency status. This item is not claimed as a closed source-internal proof unless the imported continuum and multipole ansatze are made explicit.

Imported bubble-energy ansatz 7.9 [IA]. AG81–AG86. Bubble-energy ansatz for antigravity bubbles. An antigravity bubble is a localized domain of one grav-phase embedded in a bulk region of the opposite grav-phase. For a spherical bubble of radius R, the minimal effective energy takes the form

$$E_{\mathrm{bubble}}(R) = 4\pi R^2 \tau_g + \frac{4\pi}{3} R^3 \Delta f_g + E_{\mathrm{field}}(R),$$

where τ_g is the wall tension, Δf_g is the bulk free-energy density difference, and E_{field} is the long-range field contribution.

Proposition 7.10 [D]. AG83–AG86. Bubble growth, collapse, and critical radius. *Under the bubble-energy ansatz, the bubble energy may admit shrinkage, runaway growth, or a metastable finite radius R_* depending on the sign and magnitude of Δf_g and the field contribution. A critical bubble radius separates the shrinking and growing regimes, so mixed grav-media can support nucleation-type transitions rather than only smooth crossover.*

Conditional proof sketch under the imported background and standing hypotheses. Assume IH8 and IH9. Under the bubble-energy ansatz, the sign of the bulk term and the competition between interface and field contributions determine whether the energy is monotone, metastable, or critical-radius controlled.

Dependency status. This item is not claimed as a source-internal proof unless the imported effective bubble ansatz is accepted on the stated working window.

7.4 Topological sectors and interface modes

Definition 7.11 [A]. AG87–AG90. Topological wall sectors and wall fluctuations. If the vacuum manifold of the grav-phase order parameter is disconnected,

$$\mathcal{M}_g = \{+P_0, -P_0\},$$

then domain walls are classified by $\pi_0(\mathcal{M}_g)$. For a stable wall profile $P_{\mathrm{wall}}(x)$ one studies fluctuations

$$P_g(x,t) = P_{\mathrm{wall}}(x) + \delta P(x,t),$$

and the corresponding interface-localized fluctuation branches.

Proposition 7.12 [D]. AG88–AG90. Interface protection and soft mode. *Stable grav-domain walls are protected against local smoothing that does not change the asymptotic vacua, and they carry at least one translational soft mode,*

$$\delta P(x) \propto \partial_x P_{\mathrm{wall}}(x).$$

Lean-backed witness summary. The synchronized Lean layer now records an explicit AG-T7 witness theorem. It combines the canonical domain-wall witness with an interface-branch witness from the marked-moment dictionary. In human terms, the formal layer provides source-normalized witness-level support for the claim that the AGT source admits a concrete wall sector and a concrete interface branch rather than leaving both as informal expectations.

Dependency status. Lean proves a source-normalized witness version of the existence of wall and interface-branch sectors. The stronger prose claims about full protection and the complete fluctuation spectrum still rely on the imported phase ansatz and continuum interpretation.

Theorem 7.13 [T]. AG-T7. Interface fluctuation theorem. *Stable grav-domain walls carry at least one translational zero or soft mode, at least one gapped amplitude mode, and may support interface-localized bound excitations.*

Theorem 7.14 [T]. AG-T8. Interface-sector theorem. *Suppose that the grav-phase order parameter admits at least two disconnected stable minima and that IH4, IH8, and IH9 hold on the working interface window. Then AGT admits localized interface sectors of at least four types: domain walls, grav-dipoles, neutral shells, and antigravity bubbles.*

These four object classes matter because they prevent AGT from collapsing into a bare sign-modified force law. Once they are admitted, the antigravity line has not only fields and phases but its own localized bodies.

7.5 Failure-mode register

The early AGT shell fixes four non-optional failure modes, the later field shell extends them in the expected screening direction, and the interface source block adds a second set specific to localized sectors.

1. AG-F1: grav-residue cannot be realized as a transfer-stable structural observable. In that case AGT collapses into ad hoc phenomenology.

2. AG-F2: the signed Green channel has no stable scaling window. Then repulsive local effects may occur formally, but there is no antigravity regime as a macroscopic law.

3. AG-F3: opposite-sign sectors fail to produce dynamically meaningful neutral or screened composites. Then the program loses its screening and composite-neutrality branch.

4. AG-F4: the antigravity sign survives only for a gauge-like observer artifact and disappears under admissible observer change. Then the effect is not a physical invariant of the program.

5. AG-F5 and beyond: the later source adds mesoscopic failure modes in which no stable coarse field window exists, neutral composites fail to suppress the long-range monopole channel, mixed phases run away immediately into uncontrolled segregation, or continuum closures fail to reflect the transfer-stable microscopic charge.

6. AG-F9: the effective potential has no genuinely disconnected minima. Then there are no stable grav-domain walls or topological interface sectors.

7. AG-F10: wall tension is negative or ill-defined. Then the interface description collapses before any transport or bubble physics can be trusted.

8. AG-F11: dipole and shell sectors never metastabilize. Then AGT remains only a theory of bulk phases.

9. AG-F12: bubble nucleation always runs away without a metastable window. Then the program admits only catastrophic phase conversion and loses a controlled interface sector.

These failure modes are not downstream engineering concerns. They belong to the theory statement itself and are part of what gives the AGT manuscript its audit boundary.

8 Test-Body Transport in Antigravity Backgrounds

The later AGT source does not stop at force laws and phase structure. Once signed backgrounds, screening, walls, shells, dipoles, and bubbles are admitted, the next required question is dynamical: what classes of motion are available to probe sectors in those backgrounds? This chapter isolates that transport layer so it stands on its own rather than being buried inside the prototype bridge.

8.1 Test bodies and observer-level transport

Definition 8.1 [A]. AG90–AG93. Test bodies and effective energy. A test body is a stable sector p equipped with grav-charge $q_g(p)$, inertial amplitude $M_{\mathrm{in}}(p) > 0$, and an observer transport branch $\mathcal{B}_p$. On the observer window its effective energy in a signed grav-background is

$$E_p(x,k) = E_p^{\mathrm{tr}}(k) - \Gamma_g q_g(p)\phi_g(x).$$

In the low-energy regime one expands

$$E_p^{\mathrm{tr}}(k) \approx M_{\mathrm{rest}}(p) + \frac{|k|^2}{2M_{\mathrm{in}}(p)}.$$

Proposition 8.2 [D]. AG91–AG99. Basic transport equations and central-field splitting. *Different test bodies may have the same grav-charge but different inertial amplitudes, and in the quasiclassical observer window their motion obeys*

$$\dot{x} = \nabla_k E_p(x,k), \qquad \dot{k} = -\nabla_x E_p(x,k),$$

so that in the low-energy regime

$$M_{\mathrm{in}}(p)\,\ddot{x} = \Gamma_g q_g(p)\nabla\phi_g(x).$$

For a central source with charge Q_A and a working unscreened radial potential of the form

$$\phi_A(r) \approx \kappa Q_A G_*(r), \qquad G_*'(r) < 0$$

on the observer window, same-sign products $q_g(p)Q_A > 0$ produce attraction, whereas opposite-sign products $q_g(p)Q_A < 0$ produce repulsion.

Corollary 8.3 [D]. AG99a. Coulomb/Newton window. *In the special three-dimensional Coulomb/Newton window $G_*(r) = 1/r$, the standard $1/r^2$ central split is recovered.*

Lean-backed witness summary. The synchronized Lean layer now records the witness theorem `ag\_t9\_transport\_class\_witnesses`. It explicitly realizes bound-orbit, repulsive-scattering, and screened-quasi-free central transport classes. This gives a formal witness-level closure of the central split between attractive and repulsive sectors, even though the full Hamilton-type observer expansion in the manuscript remains an effective reading rather than a fully internal derivation.

Dependency status. Lean proves a source-normalized witness version of the central transport classes. The full observer-level equations of motion in the prose statement still depend on the imported low-energy transport expansion.

The source makes an additional point that matters for AGT: repulsive sectors are not just attractive sectors with a minus sign. In the central reduction, they change the admissible trajectory class itself. The attractive channel supports bound orbits at suitable energies, whereas the repulsive channel naturally lives in scattering and expulsion sectors unless additional composite geometry or screening structure creates local traps.

8.2 Transport in structured backgrounds

The Chapter 5 source block next lists the transport backgrounds that become canonical in AGT: screened monopoles, dipoles, domain walls, neutral shells, and antigravity bubbles.

The following structured transport classes are stated under imported effective background profiles consistent with IH3, IH4, IH8, and IH9.

Proposition 8.4 [D]. AG98–AG114. Structured transport classes. *Assume IH3, IH4, IH8, and IH9. In a screened Yukawa-type medium,*

$$\phi_g(r) \sim \frac{Q}{r} e^{-m_{\mathrm{scr}} r},$$

the interaction range is finite and transport becomes quasi-free outside a screening shell. In a dipole field,

$$\phi_{\mathrm{dip}}(r) \sim \frac{\mathbf{d}_g \cdot \hat{r}}{r^2},$$

transport is anisotropic and depends on orientation relative to the dipole axis. Across a wall background, test bodies can reflect, transmit, or become temporarily trapped depending on their incoming energy. Bubble backgrounds support capture, expulsion, quasi-bound interface motion, and transporting-bubble regimes. Neutral shells suppress the far field but can remain dynamically active in a finite radial window, and grav-neutral polarizable bodies can still drift in field gradients through induced higher moments.

Lean-backed witness summary. The same AG-T9 witness theorem also provides witness-level support for explicit wall-reflection, wall-transmission, temporary-trapping, bubble-capture, and bubble-expulsion classes. The earlier composite and dipole witnesses likewise support the availability of neutral-shell and dipole sectors. The formal layer therefore provides explicit witness-level realizations of the named structured transport classes.

Dependency status. Lean proves source-normalized witness versions of the listed structured transport classes. The stronger effective-potential reading used in the prose still depends on the imported background and transport hypotheses.

Theorem 8.5 [T]. AG-T9. Test-body transport theorem. *For test bodies moving in the AGT backgrounds of central monopoles, screened media, grav-dipoles, domain walls, neutral shells, and antigravity bubbles, observer-level transport decomposes into at least the following classes: bound-orbit transport, repulsive scattering, screened quasi-free transport, anisotropic dipole scattering, wall reflection or transmission, bubble capture or expulsion, and polarizable neutral drift.*

This chapter is the first place where AGT becomes a mechanical theory of probe motion rather than only a sector-labeling or phase-classification program.

9 Spectral and Kernel Realization of Antigravity Transport

The source next closes the transport layer back onto the marked-kernel language. This is the conceptual point at which AGT stops being only an observer-level effective model and re-enters the structural core through resolvents, dominant branches, and marked moments.

9.1 Grav-marked kernel, resolvent, and marked moments

Definition 9.1 [A]. AG115–AG121. Grav-marked kernel, resolvent, and marked moments. For a fixed observer $\mathcal{O}$, the grav-marked transport kernel is

$$K_{\mathcal{O}}^{(g,\mathrm{int})}(z,\theta,\eta,s)=\sum_{e} w(e)z^{\gamma(e)}e^{i\theta\cdot\Delta x_{\mathcal{O}}(e)}e^{-\eta a(e)}s^{\Delta q_g(e)}[[e]].$$

Its resolvent is

$$\mathcal{R}_g^{\mathrm{int}}(z,\theta,\eta,s)=(I-K_{\mathcal{O}}^{(g,\mathrm{int})}(z,\theta,\eta,s))^{-1},$$

and, when a dominant branch exists, the first marked derivatives of $\log\lambda_*$ define spectral momentum, energy, and grav-charge candidates.

This subsection uses the sharpened integer-level grav-marked kernel rather than the earlier sign-level kernel of Section 5. Accordingly, the associated first marked quantity is the integer-level spectral candidate $Q_{g,\mathrm{int}}^{\mathrm{spec}}$, not the earlier sign-level candidate $Q_{g,\mathrm{sgn}}^{\mathrm{spec}}$.

Hypothesis 9.2 [H]. AG123. Working spectral compatibility hypothesis. For Sections 9–10 we impose the working hypothesis that, on stable low-energy sectors with a simple admissible dominant branch, the sign of the spectral grav-charge agrees with the sign of the microscopic grav-charge.

Theorem 9.3 [T]. AG-T10. Spectral grav-charge theorem. *Assume IH6 and the grav-marked kernel construction of Section 9.1. Then the first grav-marked moment of the dominant branch reproduces the microscopic grav-charge up to a positive scale factor on stable low-energy sectors.*

9.2 On-shell sectors, screening, and neutral composite moments

Proposition 9.4 [D]. AG124–AG137. On-shell sectors, screening, and neutral composite moments. *Assume Hypothesis 9.2. The on-shell grav-sector is read from the condition $\lambda_*=1$ or the equivalent pole condition of the grav-resolvent. Distinct positive, negative, and neutral grav-sectors can accordingly carry distinct dominant branches, rest energies, and inertial amplitudes. Screening corresponds to a dressed branch shift or gap in the low-energy expansion, and grav-neutral composites appear as sectors whose first grav-marked moment vanishes while higher marked moments need not vanish.*

Lean-backed witness summary. The synchronized Lean layer now records `ag\_t10\_grav\_marked\_moment\_matches\_charge`, which identifies the grav derivative of a dominant-branch witness with microscopic grav-charge, and it also records the neutral-composite marked-moment witness already present in `neutralComposite\_vanishing\_firstMoment`. Together these formal objects provide source-normalized witness-level support for the claim that first marked grav-moment and microscopic charge can be matched while neutral composites cancel the first moment without collapsing all higher responses.

Dependency status. Lean proves source-normalized witness versions of the marked-moment claims. The full analytic dominant-branch and dressed-resolvent reading in the prose statement still depends on Hypothesis 9.2 and the imported low-energy spectral framework.

9.3 Interfaces as branch-changing sectors

Proposition 9.5 [D]. AG138–AG142. Spectral reading of interfaces and observer shadows. *Assume Hypothesis 9.2. Domain walls and bubbles admit a spectral reading as inhomogeneous sectors in which the identity of the dominant local branch changes across an interface. In the weak-field low-energy window, the observer equations of motion arise as the shadow of dominant-branch mechanics rather than as a new independent axiom.*

Lean-backed witness summary. The synchronized Lean layer now records `ag\_t11\_wall\_dictionary\_witness` and `ag\_t12\_spectral\_transport\_witness`. The first gives an explicit spectral wall dictionary with a soft branch, amplitude data, and an interface-localized branch flag; the second gives an explicit grav-marked kernel together with a dominant-branch witness carrying momentum, energy, and grav-charge shadows. This formally supports a source-normalized witness reading of branch-changing and interface behavior.

Dependency status. Lean proves source-normalized witness versions of the interface claims and of the spectral-transport claims. The stronger claim that the full observer equations arise as low-energy shadows of a general inhomogeneous resolvent construction still depends on Hypothesis 9.2 and the imported expansion framework.

Theorem 9.6 [T]. AG-T11. Spectral interface theorem. *Stable grav-domain walls support a spectral interface description with at least one translational soft mode, at least one gapped amplitude mode, and possibly an interface-localized grav-neutral branch.*

Theorem 9.7 [T]. AG-T12. Spectral realization theorem for antigravity transport. *If microscopic grav-charge is realized by a transfer-stable marked observable, and if IH4, IH6, and IH7 hold on the working spectral and observer windows, then antigravity transport admits a spectral description in which observer-level equations of motion are low-energy spectral shadows.*

This is the structural lock of the manuscript: the transport and interface chapters cease to be merely effective once they can be read from the same spectral machinery that supports the wider program.

10 Large-Scale Mixed-Medium Consequences

Once screening, interfaces, and transport have all been admitted, the source moves to the large-scale mixed-medium regime. The canonical coarse fields are the total density,

$$\rho_{\mathrm{tot}} = \rho_+ + \rho_- + \rho_0,$$

the grav-polarization P_g, the neutral fraction ν_0, and the interface density variables that summarize wall and bubble populations.

10.1 Cosmological grav-medium and coarse closures

Definition 10.1 [A]. AG143–AG163. Cosmological grav-medium and coarse closures. A cosmological grav-medium is an observer-level coarse regime in which microscopic EFU fluctuations

are averaged out and the relevant fields are

$$\rho_+(x,t), \qquad \rho_-(x,t), \qquad \rho_0(x,t), \qquad P_g(x,t).$$

Its large-scale signed potential solves the same projected channel,

$$(L_{\mathrm{eff}} + \varepsilon)\phi_g = \rho_g, \qquad \rho_g = \rho_+ - \rho_-,$$

and its coarse dynamics is described by a two-fluid or three-fluid closure supplemented by conversion, screening, and interface-density terms.

10.2 Large-scale regimes, voids, and wall networks

Proposition 10.2 [D]. AG151–AG169. Large-scale regimes, voids, domain networks, and bubble populations. *On large scales the AGT medium admits at least the following qualitatively distinct regimes: a polarized bulk grav-medium, a grav-neutral screened medium, a repulsive void regime, a wall-dominated domain network, and a bubble or percolation regime. Repulsive voids arise when one grav-sector dynamically evacuates the observed opposite-sign component, wall networks contribute an effective stress-like term through interface tension, and long-lived bubble populations add independent coarse variables beyond the bulk densities.*

Lean-backed witness summary. The synchronized Lean layer now records an explicit AG-T15 regime witness theorem. It explicitly realizes the five named large-scale regimes: polarized bulk, screened neutral, repulsive void, wall-dominated, and bubble-percolating. The same layer also records canonical witness objects associated with AG-T13 and AG-T14. This supplies source-normalized formal support for parts of their target content without upgrading their full physical wording to closed theorems of the manuscript.

Dependency status. Lean proves source-normalized witness versions of the named fluctuation, rate-function, and regime objects. The full cosmological and hydrodynamic interpretation in the prose still depends on the imported coarse-graining and closure hypotheses.

Theorem 10.3 [T]. AG-T13 and AG-T14. Fluctuation and large-deviation targets. *In a central-limit coarse window one expects Gaussian fluctuation fields for the large-scale grav-polarization and related observables, and in a macroscopic-history window one expects a large-deviation principle for mixed-sector profiles constrained by continuity, constitutive closure, and sector-conversion terms.*

Theorem 10.4 [T]. AG-T15. Cosmological regime theorem. *If microscopic grav-charge exists and is transfer-stable, if a large-scale continuum window exists, if screening, neutralization, segregation, and interface sectors all survive coarse averaging, and if a hydrodynamic closure is valid on the chosen observer scale, then AGT admits at least five distinct large-scale regimes: polarized bulk grav-medium, grav-neutral screened medium, repulsive void regime, wall-dominated domain network, and bubble or percolation regime.*

The source is careful not to overclaim here. These large-scale regimes are part of the internal consequence map of AGT, not yet a completed cosmological fit. Their role is to show that the line has a genuine many-body and continuum extension of its own.

11 Observational Program and Forward Dependencies

The source then makes the correct move: instead of pretending that the theory is complete once the formal layers are in place, it writes a verification ladder, an observable dictionary, explicit falsifiers, and only then a prototype bridge. This chapter preserves that order.

11.1 Verification ladder and observable dictionary

Definition 11.1 [A]. AG170–AG176. Verification ladder and observable dictionary. The AGT program is checked on six successive levels: microscopic structural data, grav-marked kernels, spectral branch extraction, coarse field reconstruction, transport and phase observables, and large-scale signatures. The minimal observable dictionary includes microscopic charges and shell data, spectral grav-charge and branch data, mesoscopic fields such as ρ_g, P_g, ν_0, and ϕ_g, transport fractions such as capture, reflection, transmission, scattering, and trapping rates, and large-scale wall, void, and bubble variables.

The same source block also fixes the immediate checkpoints of the program: nontrivial microscopic grav-charge, spectral sign compatibility, neutral-composite suppression, a numerically stable repulsive window, a screening window, interface observability, and a large-scale distinction from ordinary one-component attractive media. These are not optional benchmark ideas; they are the first conditions under which AGT deserves further development.

Proposition 11.2 [D]. AG177. Internal-to-observer comparison map. *AGT remains a unified theory only if it maintains an explicit comparison chain*

$$q_g \longmapsto Q^{\mathrm{spec}}_{g,\mathrm{int}} \longmapsto \rho_g \longmapsto \phi_g \longmapsto E^{(g)} \longmapsto F^{(g)},$$

together with the corresponding composite and phase dictionaries for dipoles, polarizable neutral bodies, walls, shells, and bubbles. Here the comparison chain uses the sharpened integer-level spectral quantity $Q^{\mathrm{spec}}_{g,\mathrm{int}}$, not the earlier sign-level candidate $Q^{\mathrm{spec}}_{g,\mathrm{sgn}}$.

Lean-backed witness summary. The synchronized Lean layer records source-normalized witness theorems for several major blocks of the comparison chain, including microscopic charge, matched spectral charge, mesoscopic phases, transport classes, and the audit ladder. This provides blockwise formal support, but not a single closed end-to-end theorem for the full observer-level comparison map.

Dependency status. Lean now proves source-normalized witness versions of the main blocks of the comparison chain. The full end-to-end observer map written here still depends on the imported constructions that connect those blocks into one continuous theory statement.

Theorem 11.3 [T]. AG-T16. Falsifiability and computational realizability theorem. *AGT qualifies as a genuine physical research program only if there exists an explicit computational pipeline from structural initialization to observer-level signatures, together with finite-step checkpoints and identifiable falsifiers. Without an algorithmic schema, observer-level signatures, and clear failure conditions, the AGT line remains only a conceptual sketch.*

The corresponding falsifiers are also explicit in the source: the grav-charge may collapse to the trivial sector; the grav-marked kernel may fail to support a compatible dominant branch; repulsive transport may never stabilize; neutralization may wash out every coarse signature; mixed grav-media may be dynamically indistinguishable from ordinary attractive media; or interface sectors

may lack any transport signature at all. These are theory-level failure conditions, not public-relations cautions.

11.2 What is not claimed

This manuscript does not claim:

1. closure of AG-T1–AG-T16 in their full physical wording;
2. theorem evidence from emulator or prototype artifacts;
3. a completed phenomenological or cosmological fit;
4. a demonstrated macroscopic engineering device;
5. observer-independent antigravity outside an explicitly stated scaling window.

11.3 Prototype bridge

The AGT source does not stop at internal theory. It also contains a bridge toward toy numerics, emulator architecture, analog-transfer notes, and hardware-facing packages. The clean manuscript preserves that bridge, but it keeps the claim boundary explicit: theory does not become proved merely because a prototype package exists.

The downstream engineering line currently consists of toy numerical, emulator, and analog-transfer artifacts. These are named only as downstream consumers of the theory manuscript and provide no theorem evidence for AG-T1–AG-T16.

11.4 Lean and computational dependencies

The current Lean namespace already fixes the first AGT language, the early background-conditional Lean closures, and the later source-normalized witness theorems in the following module set:

- `GravResidue.lean`
- `SignedSource.lean`
- `CompositeSectors.lean`
- `Targets.lean`
- `Axioms.lean`
- `Theorems.lean`
- `Source/040_DomainWalls.lean`
- `Source/041_GravDipoles.lean`
- `Source/042_AntigravityBubbles.lean`
- `Source/050_TestBodies.lean`
- `Source/051_OrbitalTransport.lean`

- `Source/052_BackgroundScattering.lean`
- `Source/060_SpectralTransportKernel.lean`
- `Source/061_ResolventTransport.lean`
- `Source/062_MarkedMomentDictionary.lean`
- `Source/070_LargeScaleMedium.lean`
- `Source/071_RepulsiveVoids.lean`
- `Source/072_MixedMediumHydrodynamics.lean`

The present theory manuscript is therefore expected to stay synchronized with the following canonical objects:

- `GravSign`
- `GravResidueObservable`
- `TransferStableGravResidue`
- `SignedSourceDensity`
- `ObserverPotential`
- `ProbeInteractionEnergy`
- `EffectiveForce`
- `CompositeGravResidue`
- target wrappers for repulsive scaling and spectral observer compatibility
- source-normalized witness theorems for AG-T3–AG-T16
- the later source modules for interfaces, transport kernels, and large-scale mixed-media regimes

The synchronization rule is therefore two-level. The manuscript keeps AG-T3–AG-T16 as theory-program targets in their full physical wording, while the Lean package now proves source-normalized witness versions of those targets in `Theorems.lean`. The provenance table records that distinction explicitly so that the manuscript does not overclaim more than the formal layer actually proves.

On the computational side, the first invariants that the later AGT numerical manuscript must reproduce are the signed source construction, the attraction or repulsion regime split, the existence or failure of a repulsive scaling window, the neutral-composite or screening branch, the wall or bubble transport observables, and the large-scale mixed-medium diagnostics. In practical terms, the theory chapter fixes what the computation must mean; the computation does not retroactively redefine the theory.

11.5 Forward dependency order

The forward dependency order is therefore

grav-residue language → signed observer channel → field-regime closure → interface sectors → test-body transport → spectral closure → large-scale program → prototype bridge → acceptance criteria.

Only after this order is respected can downstream AGT artifacts be read as non-theorem implementation evidence for a particular source-faithful AGT regime, rather than as isolated software demonstrations or as proofs of AGT target statements.

Appendix: Downstream Artifact Ledger (Non-Evidential)

Downstream software, emulator, and hardware artifacts are recorded only in this appendix and are not part of theorem evidence.

This appendix is archival metadata only and may be omitted without affecting any theorem, definition, target statement, or dependency claim of the manuscript.

In this appendix, status labels mean only that a downstream artifact package exists in the recorded workspace. They do not mean theorem closure, empirical validation, or hardware demonstration.

Milestone	**Normalized status**	**Notes**
Toy line v1.1c	Recorded numerical package	Live source tree exists under legacy AGT and reports sector-selective strip and bubble resonance behavior.
P3 prototype candidate	Recorded concept package	Interface Resonance Trap / Sector-Selective Transport Emulator chosen as the first engineering route.
P3 digital architecture	Recorded architecture package	Digital/software emulator selected as the first platform-specific implementation route.
P3 software spec v0.1	Recorded specification package	MVP definition, acceptance sheet, milestone plan, and config schema are already drafted.
P3 MVP v0.1–v0.2.2	Recorded release lineage	Runnable releases exist through v0.2.2 with acceptance and recovery notes in legacy archives.
Analog and hardware transfer packages	Recorded planning packages	Analog calibration, hardware blocks, testbench, first-session, operator, and transfer packages exist as legacy artifacts.

12 Bibliography

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